Math 118: Mathematical Methods of Data Theory

Lecture 6: Tensors and the HOSVD

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Tensors

- **Scalars:** $a \in \mathbb{R}$.
- **Vectors:** One dimensional arrays of scalars $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$.
- **Matrices:** Two dimensional arrays of scalars $A = (a_{ij}) \in \mathbb{R}^{m \times n}$. Rows $=$ horizontal "slices" and columns $=$ vertical "slices" are vectors.
- **(3)-Tensors:** Three dimensional arrays of scalars $\mathcal{A} = (a_{ijk}) \in \mathbb{R}^{\ell \times m \times n}$. Slices are matrices.

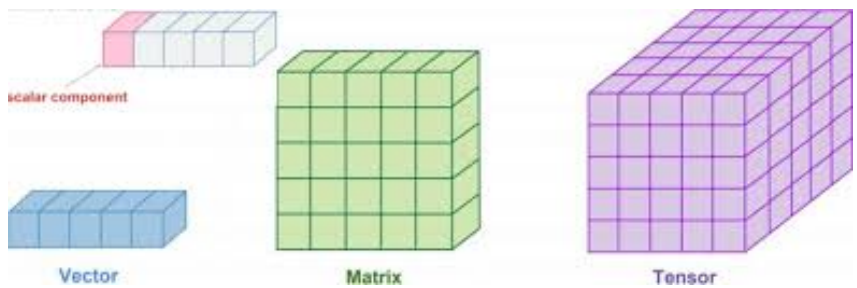


Figure: Taken from
`https://dev.to/juancarlospaco/tensors-for-busy-people-315k`

Using Tensors

- Display tensors by displaying their slices. E.g. $\mathcal{B} \in \mathbb{R}^{2\times 2\times 3}$:

$$\mathcal{B}(:,:,1) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \qquad \mathcal{B}(:,:,2) = \begin{bmatrix} -5 & 0 \\ 7 & 2 \end{bmatrix} \qquad \mathcal{B}(:,:,3) = \begin{bmatrix} 5 & 1 \\ 3 & 8 \end{bmatrix}$$

$$\mathcal{B}(1,:,:) = \begin{bmatrix} 1 & -5 & 5 \\ 3 & 0 & 1 \end{bmatrix}$$

- Will refer to each dimension as a "mode".
- Real-world uses:
  1. 1024×1024 RGB image can be stored as $\mathcal{A} \in \mathbb{R}^{1024\times 1024\times 3}$.
  2. Take collection of 64×64 greyscale images $I_1, \ldots, I_n$. Each image can be stored as a matrix $A_i \in \mathbb{R}^{64\times 64}$. Can stack these into a tensor $\mathcal{A} \in \mathbb{R}^{64\times 64\times n}$.
  3. Contingency tables ?

Tensor Algebra

- Can add to tensors if all modes are the same. E.g. if $\mathcal{B} = (b_{ijk}) \in \mathbb{R}^{2\times 2\times 3}$ and $\mathcal{C} = (c_{ijk}) \in \mathbb{R}^{2\times 2\times 3}$ then $\mathcal{B} + \mathcal{C} = (b_{ijk} + c_{ijk})$.
- **Unfolding:** Can stack slices to get matrix. Eg:

$$\mathcal{B}(:,:,1) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \qquad \mathcal{B}(:,:,2) = \begin{bmatrix} -5 & 0 \\ 7 & 2 \end{bmatrix} \qquad \mathcal{B}(:,:,3) = \begin{bmatrix} 5 & 1 \\ 3 & 8 \end{bmatrix}$$

$$\text{unfold}_1(\mathcal{B}) = \begin{bmatrix} 1 & 3 & -5 & 0 & 5 & 1 \\ 2 & 4 & 7 & 2 & 3 & 8 \end{bmatrix} \in \mathbb{R}^{2\times(2\cdot 3)} \text{ (columns become columns)}$$

$$\text{unfold}_2(\mathcal{B}) = \begin{bmatrix} 1 & 2 & -5 & 7 & 5 & 3 \\ 3 & 4 & 0 & 2 & 1 & 8 \end{bmatrix} \in \mathbb{R}^{2\times(2\cdot 3)} \textit{(rows become columns)}$$

$$\text{unfold}_3(\mathcal{B}) = \begin{bmatrix} 1 & 3 & 2 & 4 \\ -5 & 0 & 7 & 2 \\ 5 & 1 & 3 & 8 \end{bmatrix} \in \mathbb{R}^{3\times(2\cdot 2)} \textit{(3-mode vectors become columns)}$$

Tensor Algebra

Let $\mathcal{A} \in \mathbb{R}^{\ell \times m \times n}$.

- Will generalize matrix-matrix multiplication to tensor-matrix multiplication.
- **Eg:** If $A \in \mathbb{R}^{m \times n}$ have two kinds of multiplication:

$$\text{For } B \in \mathbb{R}^{n \times k} \text{ have } AB \in \mathbb{R}^{m \times k} \text{ with } (AB)_{ij} = \sum_{s=1}^{n} a_{is} b_{sj}$$

$$\text{For } B \in \mathbb{R}^{k \times m} \text{ have } BA \in \mathbb{R}^{k \times n} \text{ with } (BA)_{ij} = \sum_{s=1}^{m} a_{sj} b_{is}$$

- For 3-mode tensor, get three kinds of multiplication.

$$\text{For } B \in \mathbb{R}^{k \times \ell} \text{ have } \mathcal{A} \times_1 B \in \mathbb{R}^{k \times m \times n} \text{ with } (\mathcal{A} \times_1 B)_{ijh} = \sum_{s=1}^{\ell} b_{is} a_{sjh}$$

$$\text{For } B \in \mathbb{R}^{k \times m} \text{ have } \mathcal{A} \times_2 B \in \mathbb{R}^{\ell \times k \times n} \text{ with } (\mathcal{A} \times_2 B)_{ijh} = \sum_{s=1}^{m} b_{is} a_{ish}$$

For $B \in \mathbb{R}^{k \times n}$ have $\mathcal{A} \times_3 B \ldots$

Tensor Norms

Let $\mathcal{A}$ and $\mathcal{B} \in \mathbb{R}^{\ell \times m \times n}$.

- Tensor Frobenius Norm:

$$\|\mathcal{A}\|_F = \left(\sum_{i,j,k} a_{ijk}^2 \right)^{1/2} = \left(\sum_{i=1}^{\ell} \sum_{j=1}^{m} \sum_{k=1}^{n} a_{ijk}^2 \right)^{1/2}$$

- Tensor inner product: $\langle \mathcal{A}, \mathcal{B} \rangle = \sum_{i,j,k} a_{ijk} b_{ijk}$.
- **Note:** $\langle \mathcal{A}, \mathcal{A} \rangle = \|\mathcal{A}\|_F^2$.

The Higher Order SVD (HOSVD)

Let $\mathcal{A} \in \mathbb{R}^{\ell \times m \times n}$.

- **Recall:** For $A \in \mathbb{R}^{m \times n}$ we have the **SVD:** $A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^\top$ where U and V are orthogonal and Σ is diagonal.
- **Now have:** $\mathcal{A} = \mathcal{S} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$ where $U^{(1)}, U^{(2)}$ and $U^{(3)}$ are orthogonal and $\mathcal{S}$ is "simple".
- **Note:** $U^{(1)} \in \mathbb{R}^{\ell \times \ell}$, $U^{(2)} \in \mathbb{R}^{m \times m}$ and $U^{(3)} \in \mathbb{R}^{n \times n}$
- $U^{(1)}$ is the left singular vectors of $\text{unfold}_1(\mathcal{A}) \in \mathbb{R}^{\ell \times (m \cdot n)}$. Similar for $U^{(2)}$ and $U^{(3)}$.
- $\mathcal{S}$ has orthogonal slices:

$$\langle \mathcal{S}(i,:,:), \mathcal{S}(j,:,:) \rangle = \langle \mathcal{S}(:,i,:), \mathcal{S}(:,j,:) \rangle = \langle \mathcal{S}(:,:,i), \mathcal{S}(:,:,j) \rangle = 0 \quad \text{for } i \neq j$$

- (3-mode) **Singular values:** $\sigma_j^{(3)} = \|\mathcal{S}(:,:,j)\|_F$.

Application: (Simple) Classification Problems

Problem

- *Suppose* $I_1, \ldots, I_n$ *are greyscale handwritten digits, represented as* 64×64 *matrices.*
- *Suppose* $I_1, \ldots, I_{n_{train}}$ *are "labeled",* i.e. *know that* I_5 *is a "1",* I_{17} *is a "3".*
- **Goal:** *Find an algorithm to classify remaining unlabeled images* $I_{n_{train}+1}, \ldots, I_n$.

Possible Solution:

- Find **templates** $\tilde{I}_1, \ldots, \tilde{I}_{10} \in \mathbb{R}^{64 \times 64}$ such that $\tilde{I}_2$ represents idealized "2" and so on.
- **Specifically:** for $i = 1, \ldots, n_{\text{train}}$ want $\langle I_i, \tilde{I}_2 \rangle$ to be large if I_i is a "2" and small (or negative) otherwise.
- **Classification:** For unlabeled image I_i with $i \in \{n_{\text{train}} + 1, \ldots, n\}$ classify as "2" if $\langle I_i, \tilde{I}_2 \rangle$ is larger than $\langle I_i, \tilde{I} \rangle$ for $j \neq 2$.
- **Question:** How to find $\tilde{I}_1, \ldots, \tilde{I}_{10}$.

Using HOSVD for classification

1. Stack $I_1, \ldots, I_{n_{\text{train}}}$ into tensor $\mathcal{A} \in \mathbb{R}^{64 \times 64 \times n_{\text{train}}}$.
2. **Recall HOSVD:** $\mathcal{A} = \mathcal{S} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$ where $U^{(1)}, U^{(2)}$ and $U^{(3)}$ are orthogonal and $\mathcal{S}$ is "simple".
3. **Define:** $A_i = S(:, :, i) \times_1 \times U^{(2)} \times U^{(3)} \in \mathbb{R}^{64 \times 64}$.
4. **Claim 1:** $\mathcal{A} = \sum_{i=1}^{n_{\text{train}}} A_i \times_3 \boldsymbol{u}_i^{(3)}$ where $\boldsymbol{u}_i^{(3)}$ is the i-th column of $U^{(3)}$.
5. Use the first few[1] A_i as our templates.
6. Say A_1 represents "2"s if $\langle A_1, I_j \rangle$ is large and positive for all I_j with $j \in \{1, \ldots, n_{\text{train}}\}$ representing "2"s.

[1] In practice, we may need more than 10